MILITARY SPECIFICATION SHEET

BOLT, PROTRUDING HEAD, EXTERNAL WRENCHING, STRAIGHT SHANK, EXPANDABLE ALUMINUM ALLOY SLEEVE, ALLOY STEEL PIN, 180 KSI Ftu

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bolt described herein shall consist of this document and the latest issue of MIL-B-8831.

1. Application criteria. The bolts shall be used in connections to provide a precision net fit or controlled interference fit. When selection provides for compatibility of size, length, materials and fit, the bolts shall be used in accordance with the limitations of MIL-B-8831/1. The bolts shall not be degreased and cetyl alcohol shall not be removed before installation.

2. Companion and accessory parts. Nuts in accordance with MIL-N-8985/3 or MIL-N-8922/1 shall be used with the bolts.

3. Oversize bolts. See part number example and oversize code letter for ordering oversize shank bolts.

4. Tooling. Wrenches shall be in accordance with MIL-W-8982.

5. Military installation. Installation shall be in accordance with Navy NAVWEPS-01-1A-8 and Air Force T.O. 1-1A-8.

6. Materials.

6.1 Pin. Pin material shall be alloy steel in accordance with MIL-S-6049 (AISI 8740), MIL-S-8844, Class I (AISI 4340) or AMS 6414.

6.2 Sleeve. Sleeve material shall be in accordance with QQ-A-250/4 for aluminum alloys 2024-T3 and 2024-T42; and QQ-A-225/6 or QQ-A-200/3 for aluminum alloy 2024-T351.

7. Heat treatment.

7.1 Pin. The pin shall be heat treated in accordance with MIL-H-6875 and Rockwell hardness shall be 39-43.

7.2 Sleeve. Aluminum alloy 2024-T62 sleeve shall be heat treated in accordance with QQ-A-250/5.

FSC 5306

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8. Surface texture. Surface texture shall be R_a-32 in accordance with ANSI B46.1.

9. Coatings or surface treatment.

9.1 Pin. The pin shall be aluminum coated in accordance with Kalgard 2240C, Type I, Class A or the equivalent.

9.2 Sleeve. The inside diameter of the sleeve shall be anodized in accordance with MIL-A-8625, Type I, Class 1. The outside diameter shall not be coated.

10. Lubricants. The bolt outside diameter and sleeve inside diameter shall be lubricated with cetyl alcohol in accordance with MIL-L-87132. Dry film lubricant shall be in accordance with MIL-L-8937 with 0.0001-inch overspray on the head and threads.

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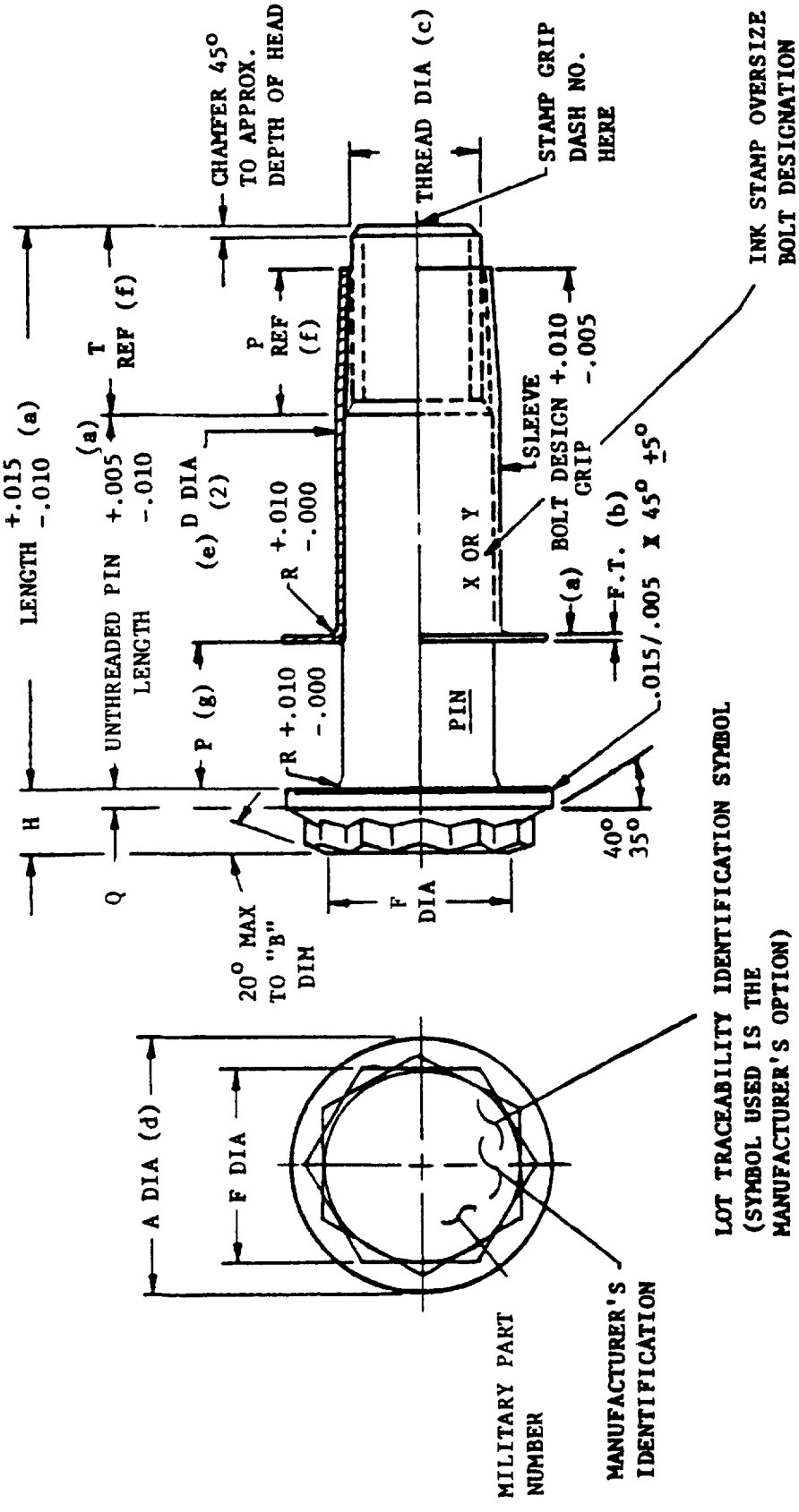


Figure 1. Flanged tension head bolt.

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NOTES:

a. Grip lengths are available in .0625 increments.

b. Dimension of non-structural flange thickness is established by sleeve bolt assembly installation requirements allowing no threads in bearing.

c. Threads shall be in accordance with MIL-S-8879. Thread major diameter (T.D.) shall be in accordance with Table I.

d. 1. For sizes .190 thru .375, dimension "A" shall be concentric to dimension "D" within .008 T.I.R.

2. For sizes .437 thru .625, .012 T.I.R.

3. For sizes .750 thru 1.000, .020 T.I.R.

e. 1. For sizes .190 thru .375, dimension "D" shall be concentric to thread pitch diameter within .006 T.I.R.

2. For sizes .437 thru .625, .010 T.I.R.

3. For sizes .750 thru 1.000, .016 T.I.R.

f. Reference dimensions are for design purposes only and are not an inspection or manufacturing requirement.

g. Protrusion will vary according to taper to provide expansion as given in Table I.

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TABLE I. Bolt dimensions.

SIZE NO'a	GROUP	GRIP DASH NUMBERS	THREAD PER MIL-S-8879 UNJF-3A	A DIA	H HEAD HEIGHT	F	F T +.015(b) -.000	Q	P (g)	R FILLET RADIUS	T D THREAD DIA	T THREAD LENGTH REF	D DIMENSION (a) (b) UNINSTALLED SLEEVE DIAMETER			D' DIMENSION (a) (b) EXPANDED SLEEVE DIAMETER			MINIMUM INSTALLATION HOLE SIZE DIAMETER		
													STD	X	Y	STD	X	Y	STD	X	Y
-3	1	-6 THRU -8	1900-32	.409	.128	.313	.010	.030	.225 .215	.025	.1840	.343	.2365 .2360	.2425 .2420	.2485 .2480	.2455 .2445	.2515 .2505	.2575 .2565	.237	.243	.249
	2	-9 THRU -16							.432 .422				.2365 .2360	.2425 .2420	.2485 .2480	.2455 .2445	.2515 .2505	.2575 .2565	.237	.243	.249
	3	-17 THRU -22		.399	.118	.305		.020	.432 .422		.1810		.2485 .2480	.2555 .2550	.2625 .2620	.2575 .2565	.2645 .2635	.2715 .2705	.249	.256	.263
-4	1	-6 THRU -8	2500-28	.516	.138	.376	.010	.045	.225 .215	.025	.2440	.406	.3005 .3000	.3115 .3110	.3175 .3170	.3095 .3085	.3205 .3195	.3265 .3255	.301	.312	.318
	2	-9 THRU -17							.432 .422				.3005 .3000	.3115 .3110	.3175 .3170	.3095 .3085	.3205 .3195	.3265 .3255	.301	.312	.318
	3	-18 THRU -28		.506	.126	.367		.035	.432 .422		.2410		.3145 .3140	.3215 .3210	.3305 .3300	.3235 .3225	.3305 .3295	.3395 .3385	.315	.322	.331
-5	1	-6 THRU -8	3125-24	.636	.173	.502	.015	.050	.225 .215	.030	.3060	.484	.3665 .3660	.3735 .3730	.3805 .3800	.3755 .3745	.3825 .3815	.3895 .3885	.367	.374	.381
	2	-9 THRU -20							.432 .422				.3665 .3660	.3735 .3730	.3805 .3800	.3755 .3745	.3825 .3815	.3895 .3885	.367	.374	.381
	3	-21 THRU -31		.624	.161	.492		.040	.432 .422		.3020		.3805 .3800	.3885 .3880	.3955 .3950	.3895 .3885	.3975 .3965	.4045 .4035	.381	.389	.396
-6	1	-6 THRU -8	3750-24	.718	.187	.564	.015	.053	.225 .215	.030	.3680	.562	.4315 .4310	.4395 .4390	.4515 .4510	.4405 .4395	.4485 .4475	.4605 .4595	.432	.440	.452
	2	-9 THRU -22							.432 .422				.4315 .4310	.4395 .4390	.4515 .4510	.4405 .4395	.4485 .4475	.4605 .4595	.432	.440	.452
	3	-23 THRU -34		.704	.175	.553		.043	.432 .422		.3640		.4475 .4470	.4555 .4550	.4635 .4630	.4565 .4555	.4645 .4635	.4725 .4715	.448	.456	.464
-7	1	-6 THRU -8	4375-20	.808	.218	.627	.020	.056	.225 .215	.040	.4310	.625	.4985 .4980	.5105 .5100	.5185 .5180	.5075 .5065	.5195 .5185	.5275 .5265	.499	.511	.519
	2	-9 THRU -26							.432 .422				.4985 .4980	.5105 .5100	.5185 .5180	.5075 .5065	.5195 .5185	.5275 .5265	.499	.511	.519
	3	-27 THRU -40		.792	.206	.616		.046	.432 .422		.4260		.5185 .5180	.5295 .5290	.5415 .5410	.5275 .5265	.5385 .5375	.5505 .5495	.519	.530	.542
-8	1	-6 THRU -8	5000-20	.891	.252	.690	.020	.060	.225 .215	.040	.4930	.703	.5695 .5690	.5795 .5790	.5895 .5890	.5785 .5775	.5885 .5875	.5985 .5975	.570	.580	.590
	2	-9 THRU -30							.432 .422				.5695 .5690	.5795 .5790	.5895 .5890	.5785 .5775	.5885 .5875	.5985 .5975	.570	.580	.590
	3	-31 THRU -46		.877	.240	.679		.050	.432 .422		.4880		.5895 .5890	.5995 .5990	.6085 .6080	.5985 .5975	.6085 .6075	.6175 .6165	.590	.600	.609

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TABLE I. Bolt dimensions. - Continued

SIZE NO'a	GROUP	GRIP DASH NUMBERS	THREAD PER MIL-S-8879 UNJF-3A	A DIA	H HEAD HEIGHT	F	F T +.015(b) -.000	Q	P (g)	R FILLET RADIUS	T D THREAD DIA	T THREAD LENGTH REF	D DIMENSION (a) (b) UNINSTALLED SLEEVE DIAMETER			D' DIMENSION (a) (b) EXPANDED SLEEVE DIAMETER			MINIMUM INSTALLATION HOLE SIZE DIAMETER		
													STD	X	Y	STD	X	Y	STD.	X	Y
-9	1	-10 THRU -23	5625-18	1.096	.280	.877	.025	.070	.432	.073	.555	.781	.6395 .6390	.6545 .6540	.6675 .6670	.6485 .6475	.6635 .6625	.6765 .6755	.640	.655	.668
	2	-24 THRU -48		1.079	.268	.865		.055	.422		.550		.6705 .6700	.6865 .6860	.7015 .7010	.6795 .6785	.6955 .6945	.7105 .7095	.671	.687	.702
	3	-49 THRU -72											.7015 .7010	.7175 .7170	.7325 .7320	.7105 .7095	.7265 .7255	.7415 .7405	.702	.718	.733
-10	1	-10 THRU -25	6250-18	1.191	.324	.940	.025	.075	.432	.073	.618	.843	.7075 .7070	.7265 .7260	.7465 .7460	.7165 .7155	.7355 .7345	.7555 .7545	.708	.727	.747
	2	-26 THRU -52		1.175	.312	.928		.060	.422		.612		.7465 .7460	.7645 .7640	.7795 .7790	.7555 .7545	.7735 .7725	.7885 .7875	.747	.765	.780
	3	-53 THRU -78											.7795 .7790	.7955 .7950	.8115 .8110	.7885 .7875	.8045 .8035	.8205 .8195	.780	.796	.812
-12	1	-10 THRU -29	7500-16	1.386	.392	1.064	.030	.080	.432	.073	.763	1.000	.8425 .8420	.8575 .8570	.8735 .8730	.8515 .8505	.8665 .8655	.8825 .8815	.843	.858	.874
	2	-30 THRU -60		1.370	.378	1.052		.065	.422		.737		.8845 .8840	.9045 .9040	.9205 .9200	.8935 .8925	.9135 .9125	.9295 .9285	.885	.905	.921
	3	-61 THRU -90											.9205 .9200	.9365 .9360	.9515 .9510	.9295 .9285	.9455 .9445	.9605 .9595	.921	.937	.952
-14	1	-15 THRU -26	8750-14	1.589	.470	1.252	.030	.090	.432 .422	.073	.868	1.156	.9825 .9820	.9985 .9980	1.0145 1.0140	.9915 .9905	1.0075 1.0065	1.0235 1.0225	.983	.999	1.015
	2	-27 THRU -67		1.573	.454	1.239	.030	.075	.900 .890		.861		.9825 .9820	.9985 .9980	1.0145 1.0140	.9915 .9905	1.0075 1.0065	1.0235 1.0225	.983	.999	1.015
	3	-68 THRU -108					.035		.900 .890				1.0455 1.0450	1.0615 1.0610	1.0765 1.0760	1.0545 1.0535	1.0705 1.0695	1.0855 1.0845	1.046	1.062	1.077
-16	1	-15 THRU -26	1.0000-12	1.788	.538	1.440	.031	.100	.432 .422	.073	.993	1.312	1.1235 1.1230	1.1395 1.1390	1.1545 1.1540	1.1325 1.1315	1.1485 1.1475	1.1635 1.1625	1.124	1.140	1.155
	2	-27 THRU -70		1.722	.522	1.427	.031	.085	.900 .890		.986		1.1235 1.1230	1.1395 1.1390	1.1545 1.1540	1.1325 1.1315	1.1485 1.1475	1.1635 1.1625	1.124	1.140	1.155
	3	-71 THRU -112					.038		.900 .890				1.1865 1.1860	1.2015 1.2010	1.2175 1.2170	1.1955 1.1945	1.2105 1.2095	1.2265 1.2255	1.187	1.202	1.218

NOTES:

(a) "D" DIA SHALL BE VERIFIED WITHIN ITS GRIP BEARING PORTION BEFORE EXPANSION, WHEN SLEEVE IS PUSHED TO PIN HEAD, THE "D" DIA SHALL BE WITHIN THE EXPANDED "D" DIA LIMITS

(b) THIS BOLT IS MANUFACTURED AS A PRECISION ASSEMBLY REMOVAL OF SLEEVE FROM PIN BY USER WILL RESULT IN LOSS OF THE BOLT'S ABILITY TO MEET MIN/MAX EXPANSION LIMITS

(c) ALL DIMENSIONS ARE IN INCHES. TOLERANCES: DECIMALS ±.010, ANGLES ±2°.

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TABLE II. Bolt weights.

GRIP NUMBER	BOLT DESIGN GRIP	-3 DIAMETER				-4 DIAMETER				-5 DIAMETER			
		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100	
				PIN	BOLT			PIN	BOLT			PIN	BOLT
5	.312	.353	.696	.8345	.8918	.353	.759	1.4816	1.5513				
6	.375	.415	.758	.8887	.9510	415	.821	1.5776	1.6581	.483	.967	3.0399	3.1729
7	438	.478	.821	.9417	1.0142	.478	.884	1.6718	1.7653	.545	1.029	3.1846	3.3349
8	.500	.540	.883	.9932	1.0767	.540	.946	1.7641	1.8714	.608	1.092	3.3792	3.5475
9	.562	.603	.946	1.0745	1.1693	603	1.009	1.8959	2.0175	.670	1.154	3.5305	3.6986
10	.625	.665	1.008	1.1300	1.2252	665	1.071	1.9934	2.1149	.733	1.217	3.6805	3.8644
11	.688	.728	1.071	1.1847	1.2898	728	1.134	2.0900	2.2240	.795	1.279	3.8294	4.0293
12	750	.790	1.133	1.2387	1.3540	790	1.196	2.1857	2.3324	.858	1.342	3.9772	4.1935
13	.812	.853	1.196	1.2920	1.4178	.853	1.259	2.2804	2.4403	.920	1.404	4.1237	4.3569
14	.875	.915	1.258	1.3446	1.4810	915	1.321	2.3741	2.5475	.983	1.467	4.2691	4.5196
15	.938	.978	1.321	1.3965	1.5439	978	1.384	2.4670	2.6541	1.045	1.529	4.4133	4.6815
16	1.000	1.040	1.383	1.4478	1.6062	1.040	1.446	2.5588	2.7601	1.108	1.592	4.5563	4.8427
17	1.062	1.103	1.446	1.6036	1.8053	1.103	1.509	2.6498	2.8656	1.170	1.654	4.6932	5.0031
18	1.125	1.165	1.508	1.6391	1.8556	1.165	1.571	2.9233	3.2000	1.233	1.717	4.8389	5.1628
19	1.188	1.228	1.571	1.7138	1.9232	1.228	1.634	3.0218	3.2834	1.295	1.779	4.9785	5.3217
20	1.250	1.290	1.633	1.7678	1.9903	1.290	1.696	3.1194	3 3974	1.358	1.842	5.4071	5.8293
21	1.312	1.353	1.696	1.8211	2.0570	1.353	1.759	3.2160	3.5108	1.420	1.904	5.5572	5.9509
22	1.375	1.415	1.758	1.8737	2.1232	1.415	1.821	3.3116	3.6236	1.483	1.967	5.7060	6.1211
23	1 438						1.884	3.4063	3.7357	1.545	2.029	5.8537	6.2905
24	1.500						1.946	3.5000	3.8473	1.608	2.092	6.0002	6.4592
25	1.562						2.009	3.5928	3.9583	1.670	2.154	6.1456	6.6271
26	1.625						2.071	3.6847	4.0686	1.733	2.217	6.2898	6.7943
27	1.688						2.134	3.7756	4.1784	1.795	2.279	6.4328	6.9607
28	1.750						2.196	3.8656	4.2875	1.858	2.342	6.5746	7.1263
29	1.812									1.920	2.404	6.7153	7.2912
30	1.875									1.983	2.467	6.8549	7.4554

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TABLE II. Bolt weights. - Continued

GRIP NUMBER	BOLT DESIGN GRIP	-6 DIAMETER				-7 DIAMETER				-8 DIAMETER			
		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100	
				PIN	BOLT			PIN	BOLT			PIN	BOLT
6	375	420	.982	4.3124	4.4538	.425	1.050	6.2633	6.4466	425	1 128	8 8419	9 0605
7	438	.483	1.045	4.5242	4.6820	.488	1.113	6.5548	6.7568	488	1 191	9.2256	9 4721
8	500	545	1.107	4.7333	4.9114	.550	1.175	6.8430	7 0684	550	1 253	9.6057	9.8807
9	562	608	1.170	5.0020	5.2015	.613	1.238	7.2025	7.4522	613	1 316	10 0672	10 3721
10	625	670	1.232	5.2190	5.4181	.675	1.300	7 5001	7 7487	675	1 378	10.4580	10 7634
11	688	733	1.295	5 4345	5.6520	738	1.363	7 7961	8 0657	738	1.441	10 8469	11 1782
12	750	795	1.357	5.6486	5.8851	.800	1.425	8.0904	8.3817	800	1.503	11 2338	11 5918
13	812	858	1.420	5 8613	6.1172	.863	1.488	8.3831	8.6966	863	1 566	11.6189	12 0042
14	875	920	1.482	6 0726	6.3485	925	1 550	8 6741	9 0104	925	1.628	12 0020	12.4154
15	938	983	1.545	6.2825	6.5788	.988	1 613	8.9634	9.3232	988	1 691	12 3833	12 8253
16	1 000	1 045	1.607	6.4909	6.8082	1.050	1 675	9.2511	9 6348	1 050	1 753	12 7627	13.2340
17	1 062	1 108	1.670	6.6980	7.0367	1.113	1 738	9 5372	9 9455	1 113	1.816	13.1401	13 6415
18	1 125	1 170	1.732	6.9037	7.2643	1.175	1.180	9.8216	10 2550	1 175	1.878	13 5157	14.0477
19	1.188	1.233	1.795	7.1000	7.4910	1.238	1.863	10.1043	10.5636	1 238	1.941	13.8894	14.4528
20	1 250	1 295	1.857	7.3109	7.7168	1.300	1.925	10.3855	10.8710	1 300	2.003	14.2613	14.8566
21	1 312	1 358	1.920	7.5124	7.9418	1.363	1.988	10.6650	11.1774	1 363	2.066	14.6313	15.2593
22	1 375	1 420	1.982	7.7125	8.1658	1.425	2.050	10.9429	11.4828	1 425	2.128	14.9994	15 6607
23	1 438	1 483	1.483	8.3234	8.8903	1.488	2.113	11.2192	11.7871	1 488	2.191	15.3656	16.0610
24	1 500	1 545	2.107	8.5375	9.0601	1.550	2.175	11 4938	12.0904	1 550	2.253	15 7300	16.4600
25	1 562	1 608	2.170	8.7501	9.2993	1.613	2.238	11 7669	12.3926	1 613	2.316	16 0925	16.8578
26	1 625	1 670	2.232	8.9614	9.5375	1.675	2 300	12.0384	12.6939	1 675	2.378	16.4532	17.2545
27	1 688	1 733	2.295	9.1712	9.7749	1.738	2.363	12.9651	13.8089	1 738	2.441	16 8121	17.6499
28	1 750	1 795	2.357	9.3797	10.0113	1.800	2.425	13.2561	14 0206	1 800	2.503	17.1691	18 0442
29	1.812	1 858	2.420	9.5867	10.2468	1.863	2.488	13.5454	14.3435	1 863	2.566	17 5243	18 4373
30	1 875	1 920	2.482	9.7924	10.4814	1.925	2.550	13.8330	14.6653	1 925	2.628	17.8777	18.8292
31	1 938	1 983	2.545	9 9966	10 7152	1.988	2.613	14.1191	14.9861	1 988	2 691	19.2192	20.3927
32	2.000	2.045	2.607	10.1995	10.9480	2.050	2.675	14.4034	15.3058	2 050	2.753	19.5988	20.6434
33	2.062	2 108	2.670	10.4010	11.1800	2.113	2.738	14 6862	15 6244	2 113	2.816	19 9765	21.0626
34	2 125	2 170	2.732	10.6011	11.4110	2.175	2.800	14.9673	15.9420	2 175	2.878	20 3524	21.4806
35	2.188					2.238	2.863	15.2467	16.2586	2 238	2 941	20 7263	21.8973
36	2.250					2.300	2.925	15.5246	16.5741	2 300	3.003	21.0984	22.3129
37	2 312					2.363	2.988	15.8009	16.8885	2 363	3.066	21.4686	22.7272
38	2.375					2.425	3.050	16.0755	17 2019	2 425	3 128	21 8369	23 140
39	2 438					2.488	3.113	16.3485	17 5143	2 488	3.191	22 2034	23 552
40	2 500					2.550	3.175	16.6199	17 8256	2 550	3.253	22 5681	23 9630
41	2 562									2 613	3 316	22 9308	24 3725
42	2 625									2 675	3 378	23 2918	24 7809
43	2 688									2 738	3 441	23 6509	25 1880
44	2 750									2 800	3 503	24 0081	25 5940
45	2 812									2 863	3 566	24.3635	25 9988
46	2.875									2 925	3.628	24.7171	26 4024

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TABLE II. Bolt weights. - Continued

GRIP NUMBER	BOLT DESIGN GRIP	-9 DIAMETER				-10 DIAMETER				-12 DIAMETER			
		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100	
				PIN	BOLT			PIN	BOLT			PIN	BOLT
10	625	.680	1.461	14.8598	15 4105	680	1.523	19.4812	20 1073	685	1 685	30 7345	31.5888
11	.688	.743	1.524	15.3300	15.9087	743	1.586	20.0636	20.7306	748	1 748	31 5773	32.4782
12	750	.805	1.586	15.7981	16.4168	805	1 648	20.6437	21.3565	.810	1 810	32 4172	33 3759
13	.812	.868	1.649	16 2641	16.9216	868	1 711	21.2214	21 9810	.873	1 873	33 2543	34.2717
14	875	.930	1.711	16 7280	17.4290	930	1 773	21 7968	22 6039	935	1 935	34 0886	35 1657
15	938	.993	1.774	17 1898	17.9331	993	1.836	22.3698	23 2254	.998	1 998	34.9201	36.0579
16	1.000	1.055	1.836	17.6495	18.4358	1 055	1.898	22.9406	23.8453	1.060	2.060	35.7489	36.9484
17	1 062	1.118	1.899	18.1072	18.9371	1 118	1.961	23.5090	24.4638	1.123	2.123	36.5748	37.8370
18	1 125	1.180	1.961	18.5628	19.4372	1 180	2.023	24.0751	25 0807	1.185	2.185	37.3979	38.7238
19	1.188	1.243	2.024	19.0163	19.9359	1 243	2.086	24.6389	25.6962	1.248	2.248	38.2183	39 6088
20	1.250	1.305	2.086	19.4677	20.4332	1 305	2.148	25.2004	26.3102	1 310	2.310	39.0358	40 4920
21	1 312	1.368	2.149	19.9171	20.9292	1 368	2.211	25.7596	26.9226	1.373	2.373	39 8507	41.3734
22	1 375	1.430	2.211	20 3644	21 4239	1 430	2.273	26.3165	27.5337	1.435	2.435	40 6627	42.2530
23	1 438	1.493	2.274	20 8097	21.9173	1 493	2.336	26.8711	28.1432	1.498	2.498	41.4720	43.1309
24	1 500	1.555	2.336	22.5426	23.9572	1 555	2.398	27.4235	28 7512	1.560	2.560	42 2785	44.0069
25	1 562	1.618	2.399	23.0341	24.2806	1 618	2.461	27.9735	29 3578	1.623	2.623	43.0822	44 8812
26	1 625	1.680	2.461	23.5235	24.8227	1 680	2.523	30.1966	32 0762	1.685	2.685	43.8832	45.7537
27	1.688	1.743	2.524	24.0108	25.3634	1 743	2.586	30.8028	32.4606	1.748	2.748	44.6815	46 6245
28	1 750	1.805	2.586	24.4959	25.9027	1 805	2.648	31.4065	33.1300	1.810	2.810	45.4770	47.4934
29	1.812	1.868	2.649	24.9789	26.4406	1 868	2.711	32.0079	33.7979	1 873	2.873	46.2698	48.3606
30	1 875	1.930	2.711	25.4598	26.9772	1 930	2.773	32.6069	34.4643	1.935	2 935	49.7206	52.4961
31	1 938	1.993	2.774	25.9385	27.5123	1 993	2.836	33.2035	35.1291	1.998	2.998	50 5916	52.9939
32	2.000	2.055	2.836	26.4151	28.0461	2 055	2.898	33.7977	35.7924	2.060	3.060	51.4598	53.9457
33	2 062	2.118	2.899	26.8896	28.5786	2 118	2.961	34.3896	36.4541	2.123	3.123	52.3251	54.8957
34	2 125	2.180	2.961	27.3620	29.1097	2 180	3.023	34.9791	37.1144	2.185	3.185	53.1875	55 8439
35	2 188	2.243	3.024	27.8323	29.6394	2 243	3.086	35.5663	37.7731	2.248	3 248	54.0471	56 7902
36	2 250	2.305	3.086	28.3004	30.1677	2 305	3.148	36.1511	38.4303	2.310	3.310	54.9039	57.7347
37	2 312	2.368	3.149	28.7665	30.6947	2 368	3.211	36.7336	39.0860	2.373	3.373	55 7578	58.6773
38	2 375	2.430	3.211	29.2305	31 2204	2 430	3.273	37.3137	39.7401	2.435	3.435	56 6089	59 6181
39	2 438	2.493	3.274	29.6924	31 7447	2 493	3.336	37.8915	40.3928	2.498	3.498	57 4571	60.5571
40	2 500	2.555	3.336	30.1522	32.2676	2 555	3.398	38.4670	41.0440	2.560	3.560	58 3026	61.4943
41	2 562	2.618	3.399	30.6099	32.7893	2 618	3.461	39.0401	41.6936	2.623	3.623	59.1452	62.4296
42	2 625	2.680	3.461	31.0656	33.3095	2 680	3 523	39.6109	42.3418	2.685	3.685	59 9850	63 3632
43	2 688	2.743	3.524	31.5192	33.8285	2 743	3.586	40 1794	42.9884	2.748	3.748	60.8219	64.2949
44	2 750	2.805	3.586	31.9707	34.3461	2 805	3.648	40.7456	43.6336	2.810	3.810	61.6561	65.2247
45	2 812	2.868	3.649	32 4202	34.8623	2 868	3.711	41.3095	44.2773	2.873	3.873	62.4875	66.1528
46	2.875	2.930	3 711	32.8676	35.3773	2 930	3 773	41.8711	44.9195	2 935	3 935	63 3161	67.0791
47	2 938	2.993	3.774	33.3129	35.8909	2 993	3.836	42.4303	45 5602	2 998	3 998	64 1419	68.0036
48	3 000	3.055	3.836	33.7562	36.4032	3 055	3.898	42.9873	46.1994	3 060	4 060	64.9649	68 9262
49	3 062	3.118	3.899	36.8308	40 1148	3 118	3.961	43 5420	46.8371	3.123	4 123	65 7851	69.8471
50	3 125	3.180	3.961	37.3199	40 2244	3 180	4.023	44.0944	47.4734	3 185	4.185	66 6025	70 7662
51	3 188	3.243	4 024	37.8070	40.7861	3 243	4 086	44 6446	48 1082	3.248	4.248	67 4172	71 6835
52	3 250	3.305	4 086	38.2918	41 3463	3.305	4 148	45.1924	48 7415	3 310	4.310	68 2291	72.5990
53	3 312	3.368	4 149	38.7746	41 9053	3 368	4.211	49.1791	53.5271	3 373	4 373	69 0382	73.5127
54	3 375	3.430	4.211	39.2552	42.4628	3 430	4 273	49.7829	53.6281	3 435	4 435	69 8446	74.4246
55	3 438	3.493	4.274	39 7337	43.0189	3 493	4.336	50.3843	54.3210	3.498	4 498	70.6482	75.3348
56	3 500	3.555	4.336	40.2101	43 5737	3 555	4 398	50.9834	55 0124	3.560	4 560	71.4491	76.2432
57	3 562	3.618	4.399	40.6843	44.1271	3 618	4.461	51.5801	55.7023	3 623	4.623	72 2473	77.1498
58	3 625	3.680	4.461	41.1564	44.6792	3 680	4 523	52 1744	56.3907	3.685	4.685	73.0427	78.0546
59	3 688	3.743	4.524	41 6265	45 2299	3 743	4 586	52 7664	57 0775	3 748	4 748	73.8353	78 9577
60	3 750	3.805	4.586	42.0944	45 7792	3 805	4.648	53.3560	57.7627	3.810	4.810	74.6252	79 8590
61	3 812	3.868	4.649	42.5602	46.3272	3 868	4 711	53.9432	58 4465	3.873	4.873	80.8780	87.2144
62	3 875	3 930	4.711	43.0240	46.8739	3 930	4 773	54.5281	59.1288	3 935	4.935	81.7462	87 2549
63	3 938	3.993	4.774	43.4856	47 4191	3 993	4.836	55.1107	59.8095	3 998	4.998	82.6116	88.2371
64	4 000	4.055	4.836	43.9452	47 9631	4 055	4.898	55.6909	60.4887	4.060	5 060	83.4741	89.2176
65	4 062	4.118	4.899	44.4027	48.5057	4 118	4.961	56.2688	61.1664	4.123	5.123	84.3338	90.1961
66	4 125	4.180	4.961	44.8581	49.0469	4 180	5.023	56 8443	61 8426	4.185	5 185	85 1907	91.1728
67	4 188	4.243	5 024	45.3115	49 5869	4 243	5.086	57 4175	62 5173	4 248	5 248	86.0447	92 1477
68	4 250	4.305	5.086	45 7628	50.1254	4 305	5.148	57.9884	63.1905	4.310	5.310	86.8958	93 1208
69	4 312	4.368	5.149	46.2120	50.6627	4 368	5.211	58.5570	63 8623	4 373	5.373	87 7442	94.0920
70	4 375	4 430	5 211	46.6592	51.1986	4 430	5.273	59.1233	64.5325	4 435	5 435	88.5897	95.0615
71	4 438	4 493	5.274	47.1043	51 7332	4 493	5.336	59 6873	65 2012	4 498	5 498	89 4324	96 0290
72	4 500	4.555	5.336	47.5474	52 2665	4 555	5 398	60 2489	65 8684	4 560	5 560	90 2722	96 9948
73	4 562					4 618	5.461	60.8083	66.5342	4 623	5 623	91 1093	97.9588
74	4 625					4 680	5 523	61.3653	67.1985	4 685	5 685	91 9436	98 9209
75	4 688					4 743	5.586	61 9201	67.8612	4 748	5 748	92 7750	99.8812
76	4 750					4 805	5.648	62.4726	68 5226	4.810	5 810	93.6037	100.8398
77	4 812					4 868	5.711	63.0228	69.1824	4.873	5.873	94.4296	101.7965
78	4 875					4 930	5.773	65.5708	69.8408	4.935	5.935	95.2526	102 7514
79	4 938									4 998	5 998	96 0729	103.7045
80	5 000									5.060	6 060	96.8905	104 6559
81	5 062									5 123	6 123	97 7052	105.6054
82	5 125									5 185	6 185	98 5172	106 5531
83	5 188									5 248	6 248	99.3264	107.4991
84	5 250									5.310	6 310	100 1329	108.4433
85	5 312									5 373	6 373	100.9366	109 3857
86	5 375									5 435	6 435	101 7375	110 3263
87	5 438									5 498	6 498	102.5357	111 2652
88	5 500									5 560	6.560	103 3312	112 2023
89	5 562									5.623	6 623	104 1239	113 1376
90	5 625									5 685	6 685	104 9140	114.0711

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TABLE II. Bolt weights. - Continued

GRIP NUMBER	BOLT DESIGN GRIP	-14 DIAMETER				-16 DIAMETER			
		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100	
				PIN	BOLT			PIN	BOLT
15	.938	.998	2.154	53.3394	54.8063	.999	2.311	76.5901	78.5074
16	1.000	1.060	2.216	54.4806	56.0077	1.062	2.373	78.0861	80.0525
17	1.062	1.123	2.279	55.6185	57.2245	1.124	2.436	79.5784	81.6452
18	1.125	1.185	2 341	56.7532	58.4391	1 186	2.498	81.0669	83.2354
19	1.188	1.248	2.404	57.8846	59.6517	1.249	2.561	82.5517	84 8233
20	1.250	1.310	2.466	59.0127	60.8621	1.311	2.623	84.0328	86.4087
21	1.312	1.373	2.529	60.1376	62.0705	1.374	2.686	85.5101	87.9917
22	1.375	1.435	2.591	61.2592	63.2767	1.436	2.748	86.9837	89.5723
23	1 438	1.498	2.654	62.3776	64.4809	1 499	2.811	88.4535	91.1505
24	1.500	1.560	2.716	63.4927	65.6829	1.561	2 873	89.9197	92 7263
25	1.562	1.623	2.779	64.6046	66.8829	1.624	2.936	91.3821	94.2997
26	1.625	1.685	2.841	65.7133	68.0807	1 686	2.998	92.8408	95.8707
27	1.688	1.748	2.904	67.4881	69.9459	1.749	3.061	95.0628	98.2062
28	1 750	1.810	2.966	68.6385	70 9510	1 811	3.123	96.5693	99.5566
29	1.812	1.873	3.029	69 7873	72 1754	1 874	3 186	98 0741	101.1580
30	1.875	1.935	3.091	70.9345	73.3988	1.936	3.248	99.5771	102 7583
31	1.938	1.998	3.154	72.0802	74.6212	1.999	3.311	101.0783	104.3574
32	2.000	2.060	3.216	73.2243	75.8426	2.061	3.373	102.5776	105.9553
33	2.062	2.123	3.279	74.3668	77.0629	2 124	3.436	104 0752	107.5521
34	2.125	2.185	3 341	75 5078	78 2822	2 186	3.498	105 5709	109.1477
35	2.188	2.248	3.404	76 6471	79 5005	2 249	3.561	107.0649	110 7421
36	2.250	2.310	3.466	77.7849	80.7178	2 311	3.623	108 5570	112.3354
37	2.312	2.373	3.529	78 9212	81.9341	2.374	3.686	110.0474	113.9275
38	2.375	2.435	3.591	80.0558	83.1493	2 436	3.748	111.5359	115.5185
39	2.438	2.498	3 654	81 1889	84.3636	2.499	3.811	113.0227	117.1083
40	2 500	2 560	3 716	82 3204	85 5768	2 561	3 873	114 5077	118 6969
41	2.562	2.623	3.779	83.4504	86.7890	2.624	3 936	115 9908	120 2844
42	2.625	2.685	3.841	84 5788	88.0002	2 686	3.998	117 4722	121.8707
43	2.688	2.748	3.904	85.7056	89 2104	2 749	4 061	118 9518	123 4559
44	2 750	2.810	3.966	86 8309	90 4196	2 811	4 123	120.4296	125 0398
45	2 812	2.873	4 029	87 9546	91 6277	2 874	4 186	121 9056	126 6227
46	2.875	2.935	4 091	89.0767	92.8349	2 936	4 248	123 3798	128.2044
47	2.938	2.998	4.154	90.1973	94.0410	2 999	4 311	124.8522	129 7849
48	3.000	3.060	4.216	91.3163	95 2462	3 061	4 373	126.3228	131 3643
49	3.062	3.123	4.279	92.4338	96 4503	3 124	4 436	127.7917	132 9425
50	3 125	3 185	4.341	93 5497	97 6535	3 186	4 498	129.2588	134 5195
51	3.188	3.248	4 404	94 6640	98 8556	3.249	4 561	130 7240	136 0954
52	3.250	3.310	4 466	95.7768	100.0567	3 311	4 623	132 1875	137 6702
53	3 312	3 373	4 529	96.8881	101 2568	3 374	4 686	133 6493	139 2438
54	3.375	3 435	4 591	97 9977	102 4559	3 436	4 748	135 1092	140 8163
55	3 438	3.498	4 654	99 1059	103 6540	3 499	4 811	136 5673	142 3876
56	3 500	3 560	4 716	100 2125	104 8511	3 561	4 873	138 0237	143 9577
57	3 562	3 623	4 779	101 3175	106 0473	3 624	4 936	139 4783	145 5267
58	3 625	3 685	4 841	102 4210	107 2424	3 686	4 998	140 9311	147 0946
59	3.688	3.748	4 904	103 5229	108 4365	3 749	5 061	142 3822	148 6613
60	3.750	3.810	4.966	104 6233	109 6296	3 811	5 123	143 8315	150 2268
61	3 812	3.873	5 029	105 7222	110 8217	3 874	5 186	145 2790	151 7913
62	3.875	3 935	5 091	106 8195	112 0128	3 936	5 248	146 7247	153 3545
63	3 938	3.998	5.154	107 9152	113 2029	3 999	5 311	148 1686	154 9166
64	4 000	4.060	5.216	109 0095	114 3920	4 061	5 373	149 6108	156 4776
65	4 062	4.123	5 279	110 1021	115 5801	4 124	5 436	151 0513	158 0375
66	4.125	4.185	5 341	111 1933	116 7673	4 186	5 498	152.4899	159 5962
67	4.188	4.248	5 404	112 2829	117 9534	4.249	5 561	153 9268	161 1537
68	4 250	4.315	5 471	121 7849	128 4466	4 311	5 623	155 3619	162.7101
69	4.312	4.378	5 534	122 9891	130 3210	4 374	5 686	156 7953	164 2654
70	4.375	4.440	5 596	124 1917	131 6433	4 436	5 748	158 2269	165 8195
71	4 438	4.503	5 659	125.3926	132.9647	4 506	5 818	170.8843	181 8663
72	4 500	4.565	5.721	126 5919	134.2849	4 568	5 880	172.4624	181.6609
73	4 562	4 628	5 784	127 7897	135.6042	4 631	5 943	174 0386	183 3805
74	4 625	4 690	5 846	128 9858	136 9224	4 693	6 005	175 6130	185 0988
75	4 688	4 753	5 909	130 1803	138 2395	4 756	6 068	177 1856	186 8160

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TABLE II. Bolt weights. - Continued

GRIP NUMBER	BOLT DESIGN GRIP	-14 DIAMETER				-16 DIAMETER			
		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100	
				PIN	BOLT			PIN	BOLT
76	4.750	4.815	5.971	131.3732	139.5556	4.818	6.130	178.7563	188.5320
77	4.812	4.878	6.034	132.5644	140.8707	4.881	6.193	180.3251	190.2468
78	4.875	4.940	6.096	133.7541	142.1847	4.943	6.255	181.8921	191.9604
79	4.938	5.003	6.159	134.9422	143.4977	5.006	6.318	183.4573	193.6728
80	5.000	5.065	6.221	136.1286	144.8097	5.068	6.380	185.0206	195.3840
81	5.062	5.128	6.284	137.3135	146.1206	5.131	6.443	186.5821	197.0940
82	5.125	5.190	6.346	138.4967	147.4305	5.193	6.505	188.1418	198.8029
83	5.188	5.253	6.409	139.6784	148.7393	5.256	6.568	189.6996	200 5105
84	5.250	5.315	6.471	140.8584	150.0472	5.318	6.630	191.2555	202 2170
85	5.312	5.378	6.534	142.0369	151.3539	5.381	6.693	192.8097	203.9222
86	5.375	5.440	6.596	143.2137	152.6597	5.443	6.755	194.3620	205 6263
87	5.438	5.503	6.659	144.3898	153.9644	5.506	6.818	195.9124	207.3292
88	5.500	5.565	6.721	145.5626	155.2681	5.568	6.880	197.4611	209 0309
89	5.562	5.628	6.784	146.7347	156.5707	5.631	6 943	199 0078	210.7315
90	5.625	5.690	6.846	147.9052	157.8723	5.693	7.005	200.5528	212 4308
91	5.688	5.753	6.909	149.0740	159.1729	5.756	7.068	202.0959	214 1290
92	5.750	5.815	6.971	150.2413	160.4725	5.818	7.130	203.6372	215 8259
93	5.812	5.878	7.034	151.4070	161.7710	5.881	7.193	205.1767	217 5217
94	5.875	5 940	7.096	152.5711	163.0685	5.943	7.255	206.7144	219 2163
95	5.938	6.003	7.159	153.7336	164.3650	6.006	7.318	208 2502	220 9098
96	6.000	6.065	7.221	154.8946	165.6605	6.068	7.380	209 7842	222.6020
97	6.062	6.128	7.284	156.0539	166.9549	6.131	7.443	211.3163	224.2931
98	6.125	6.190	7.346	157.2116	168.2483	6.193	7.505	212.8467	225 9830
99	6.188	6.253	7.409	158.3678	169.5406	6.256	7.568	214 3752	227.6717
100	6.250	6.315	7.471	159.5224	170.8320	6.318	7.630	215.9019	229 3593
101	6 312	6.378	7.534	160.6754	172.1223	6 381	7.693	217 4268	231.0456
102	6.375	6 440	7 596	161.8268	173.4116	6.443	7.755	218 9499	232 7308
103	6.438	6.503	7.659	162.9766	174.6999	6.506	7.818	220.4711	234 4148
104	6.500	6.565	7.721	164.1249	175.9872	6.658	7.880	221.9905	236.0977
105	6.562	6.628	7.784	165.2716	177.2734	6.631	7.943	223.5081	237 7793
106	6.625	6.690	7.846	166.4167	178.5586	6.693	8.005	225.0239	239 4598
107	6.688	6.753	7.909	167.5602	179.8429	6.756	8.068	226.5379	241 1391
108	6 750	6.815	7 971	168.7022	181.1260	6.818	8.130	228 0501	242 8173
109	6.812					6.881	8.193	229.5605	244 4943
110	6.875					6.943	8.255	231.0690	246.170.
111	6.938					7.006	8.318	232.5758	247 844
112	7.000					7.068	8.380	234.0807	249 5132
113	7.062					7.131	8.443	235.5838	251.1905
114	7 125					7 193	8.505	237.0852	252 86.6
115	7.188					7.256	8.568	238.5847	254 5316
116	7.250					7.318	8.630	240.0824	256.2004
117	7.312					7.381	8.693	241.5783	257.8680
118	7.375					7.443	8.755	243.0724	259.5345
119	7.438					7.506	8.818	244.5647	261.1998
120	7.500					7.568	8.880	246.0552	262 8639
121	7.562					7.631	8.943	247 5440	264 5269
122	7.625					7.693	9.005	249.0309	266 1887
123	7.688					7.756	9.068	250.5.60	267.8494
124	7.750					7.818	9.130	251.9993	269 5089
125	7.812					7.881	9.193	253.4808	271 1672
126	7.875					7.943	9.255	254.9606	272 8244
127	7.938					8.006	9.318	256.4385	274.4804
128	8.000					8.068	9.380	257.9147	276 1353

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TABLE III. Mechanical test requirements.

Dia Dash Number	Ultimate Tensile Strength Lbs-Min (b)	Thread Area Inches^2 (b)	Double Shear Strength Lbs-Min (a)	Pin Test Dia. (±.0005) (a)	Pin Test Dia. Area (a)	Max. Install. Torque In-Lbs	Fatigue Test Loading Lbs ±2% (c)	
							Max	Min
-3	3,910	.0217	6,100	.1900	.0284	60	1,800	180
-4	6,980	.0388	10,600	.2500	.0491	115	3,210	321
-5	11,100	.0614	16,600	.3125	.0767	185	5,100	510
-6	17,100	.0950	23,900	.3750	.1104	335	7,860	786
-7	23,200	.1288	32,500	.4375	.1503	540	10,650	1,065
-8	30,900	.1717	42,400	.5000	.1963	805	14,200	1,420
-9	39,200	.2176	53,700	.5625	.2485	1080	18,000	1,800
-10	49,000	.2724	66,300	.6250	.3068	1560	22,500	2,250
-12	71,000	.3952	95,400	.7500	.4418	2930	32,700	3,270
-14	97,100	.5392	129,900	.8750	.6013	4340	44,600	4,460
-16	126,000	.7027	169,600	1.000	.7854	6100	57,900	5,790

Notes:

a. Shear test performed on pins ground to pin test diameter. The shear values are based on a minimum single shear strength of 108 KSI and the stress area at the test diameter.

b. The ultimate tensile strength values are calculated on the basic pitch diameter of the external thread.

c. Maximum fatigue load is based on 46% of the ultimate tensile strength. Minimum fatigue load is 10% of maximum fatigue load.

11. Example of military part number.

M8831/3 - 12 X 34

- 34 — Grip length dash number in sixteenths of an inch. (Stamp on threaded end.)
- X — Oversize shank diameter. (X) = 1st oversize, (Y) = 2nd oversize. (Stamp on sleeve.)
- 12 — Diameter dash number in sixteenths of an inch.
- M8831/3 — Basic specification sheet number.

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12. Administrative and contractual provisions.

a. For design purposes and in case of conflict, this document takes precedence over documents referenced herein.

b. Definitions of terms used in this document are in accordance with MIL-STD-1515.

c. For additional specification sheets and standards, see MIL-B-8831, Supplement 1.

d. The bolt assemblies identified herein are proprietary products of P.B. Fasteners and Deutsch Fastener Corporation and are covered by the following patents: U.S. 4,087,896 (expiring May 9, 1997); 4,102,036 (expiring July 25, 1997); 4,048,898 (expiring September 20, 1996); Great Britain 1,524,886; Canada 1,023,585; Japan 50-125256; Italy 1,051,593; Israel 515578; and Germany 2,546,641.5. Additional patents are pending in France and Switzerland. The Government does not have a royalty-free license.

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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-B-8831/3 Bolt, Protruding Head, External Wrenching, Straight Shank, Expandable Aluminum Alloy Sleeve, Alloy Steel Pin, 180 KSI Ftu

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